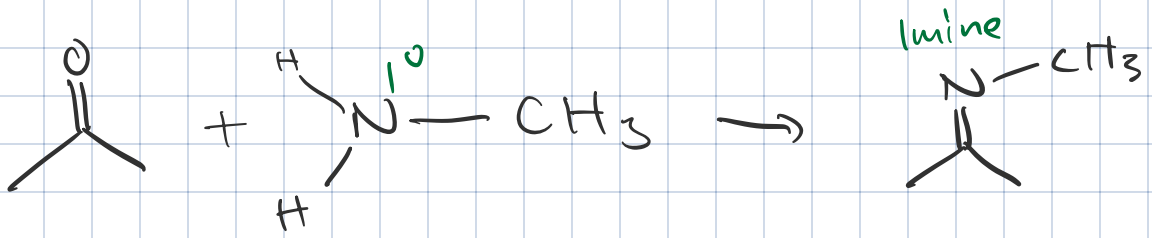
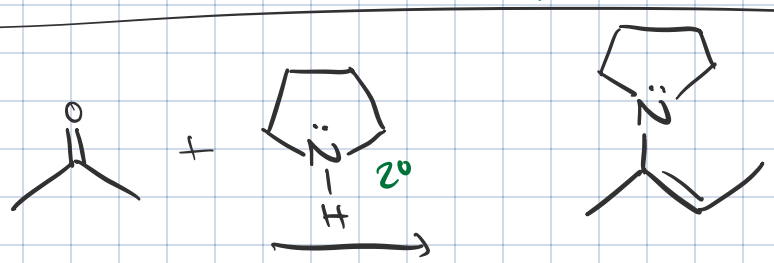
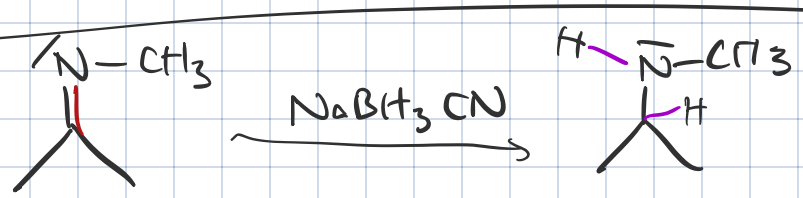


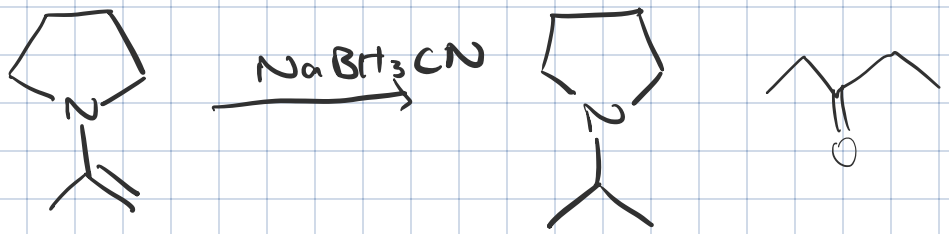
Aldehyde/Ketone + 1° amine (or 0°) → imine



Aldehyde/Ketone + 2° amine → enamine

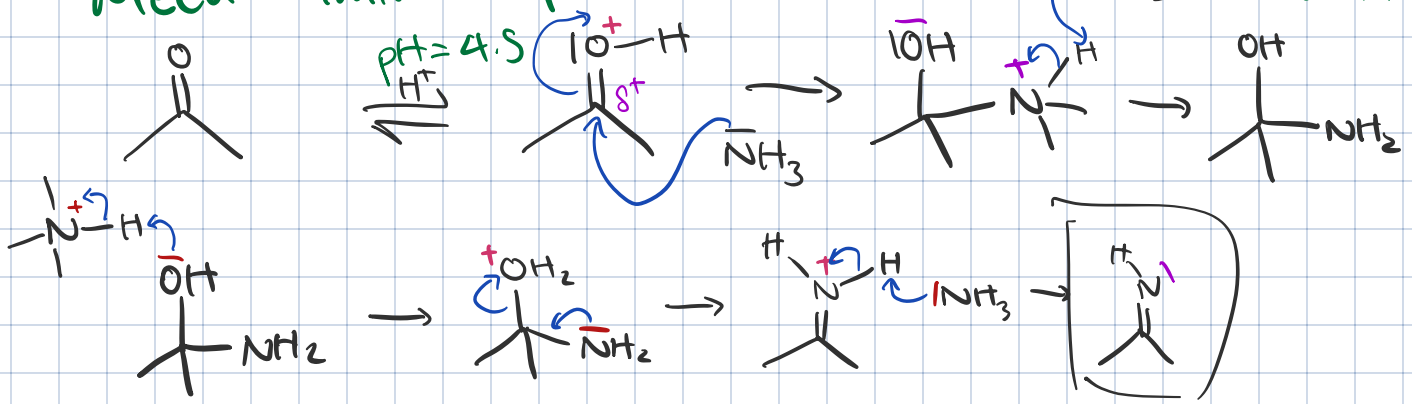


imine → reductive amination to 2° amine

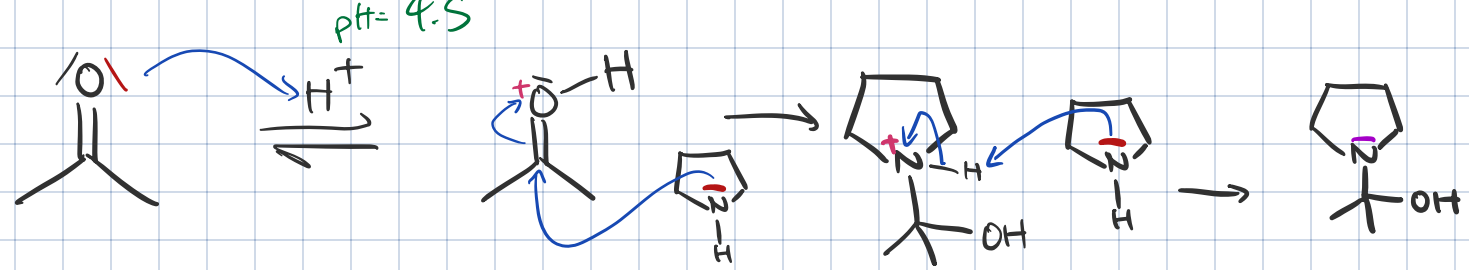


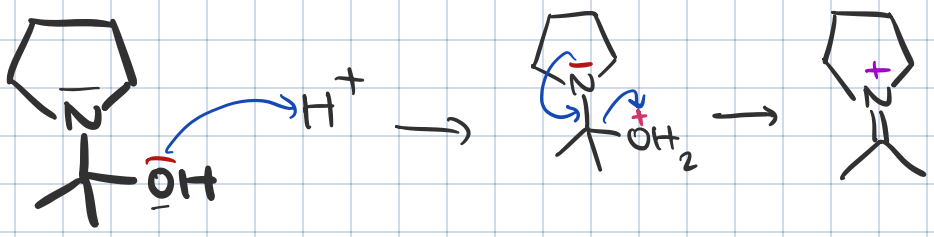
enamine → reductive amination to a 3° amine

Mech Imine formation

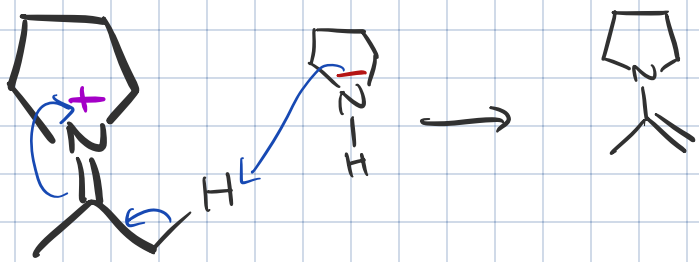


Enamine Formation



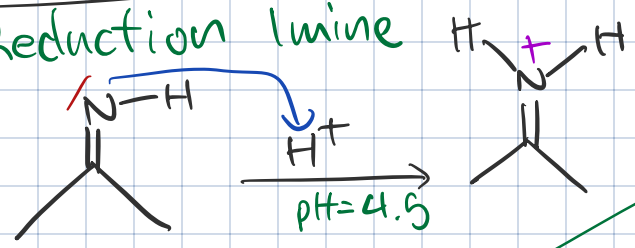


This is where the 2 mechanisms become different because the N doesn't have a H, so we can't deprotonate like we did for the imine

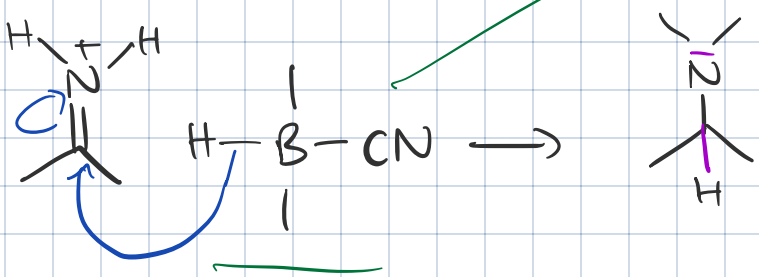


Reduction Imine

1.



2.

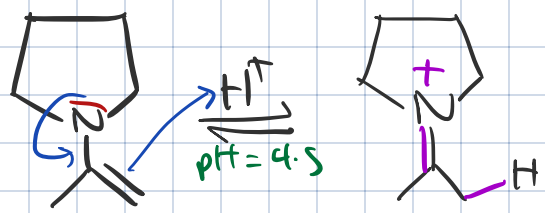


Why $NaBH_3CN$ instead of $NaBH_4$

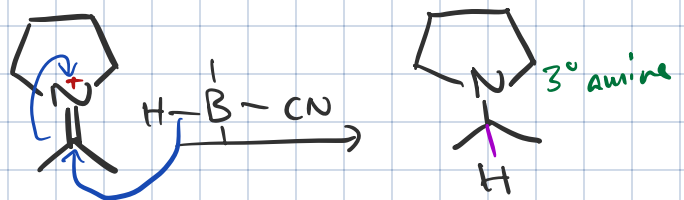
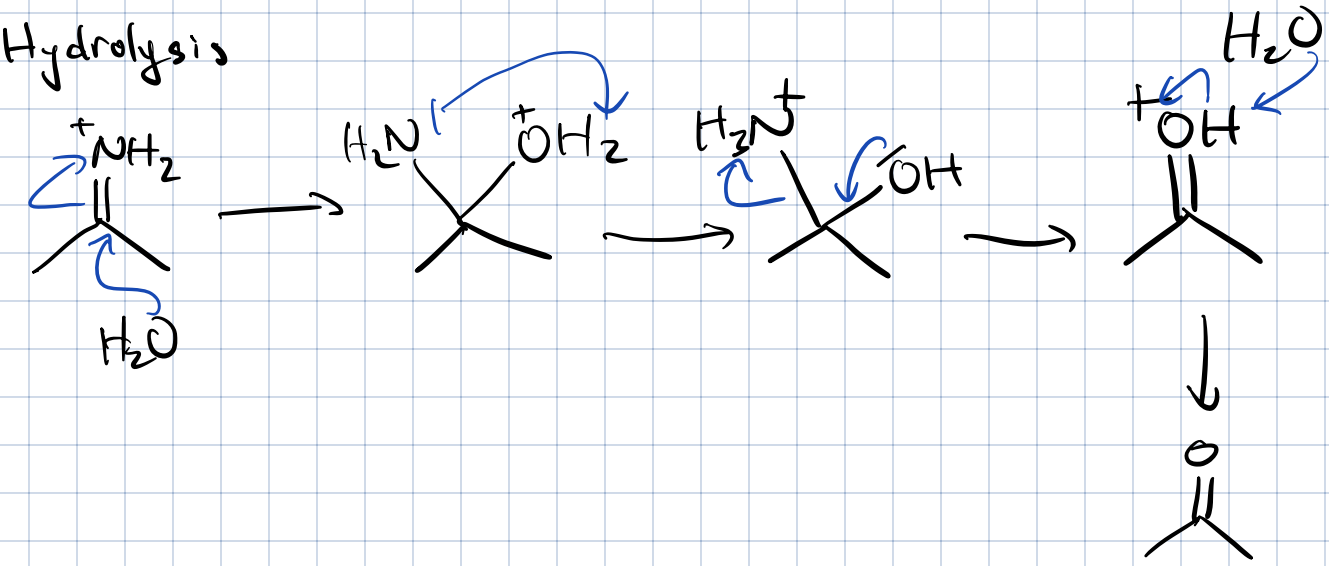
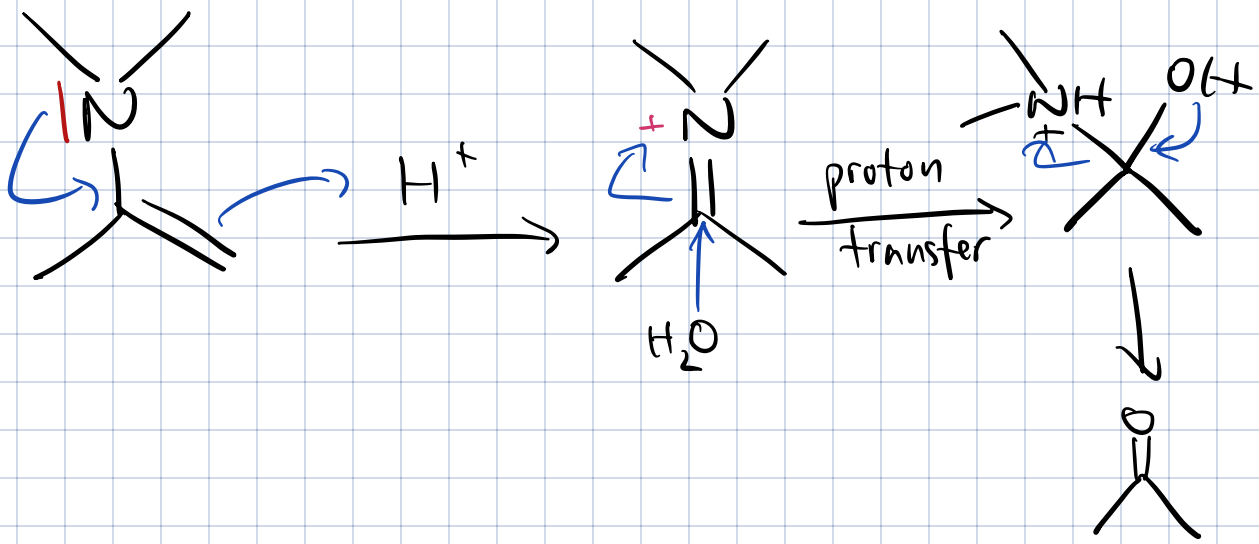
1. Acidic conditions, CN able strongly withdraw e^- from hydrogens, so they won't react to form $H_2(g)$
2. Chemiselectivity

Reduction Enamine

1.



2.



Hydrolysis